F18CF - Assignment 1

Instructions:

- This assignment is due at 23:59 (Edinburgh Time) on the date specified above.
- Your solutions can be written or typeset. No paper submission is necessary.
- *Gradescope* submission link available on assignment page in Canvas.
- The assessment criteria are listed on the assignment page.
- Start each question on a new page (Parts of the same question can be on the same page).
- This assignment contains 4 questions worth a total of 40 marks.
- Marks are listed with each question.
- An additional 3 mark is awarded for meeting the submission standards.

Questions:

8 marks

1. (a) Let $S = \{(2, a, -3, 1)^\top, (4, 3, b, -5)^\top\}$. For each of the following vectors, find the values of a and b such that the following vectors are in span(S):
   i. $\vec{u} = (4, 7, 15, -19)^\top$
   ii. $\vec{v} = (2a, 4, 9a, -20)^\top$

Solution: To be included in the span, we need $\alpha(2, a, -3, 1)^\top + \beta(4, 3, b, -5)^\top = \vec{u}$ for some values of α, β. Converting to an augmented matrix gives:

$$\left(\begin{array}{cc|c} 2 & 4 & 4 \\ a & 3 & 7 \\ -3 & b & 15 \\ 1 & -5 & -19 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 2 & 2 \\ 0 & 7 & 21 \\ 0 & 3-2a & 7-2a \\ 0 & b+6 & 21 \end{array}\right)$$

Row 2 tells us $3\beta = 21$. So for this to be consistent, rows 3 and 4 must satisfy:

$$3(3-2a) = 7-2a \implies a = \frac{1}{2}$$
$$3(b+6) = 21 \implies b = 1$$

Similarly for $\vec{v}$, we need $\alpha(2, a, -3, 1)^\top + \beta(4, 3, b, -5)^\top = \vec{v}$ for some values of α, β. Converting to an augmented matrix gives:

$$\left(\begin{array}{cc|c} 2 & 4 & 2a \\ a & 3 & 4 \\ -3 & b & 9a \\ 1 & -5 & -20 \end{array}\right) \sim \left(\begin{array}{cc|c} 1 & 2 & a \\ 0 & 7 & a+20 \\ 0 & 3-2a & 4-a^2 \\ 0 & b+6 & 12a \end{array}\right)$$

Row 2 tells us $7\beta = a + 20$. So for this to be consistent, rows 3 and 4 must satisfy:

$$\frac{a+20}{7} = \frac{4-a^2}{3-2a} \implies 5a^2 - 37a + 32 = 0 \implies a = 1, \frac{32}{5}$$
$$\frac{a+20}{7} = \frac{12a}{b+6} \implies b = \frac{78a-120}{a+20} \implies b = -2, \frac{158}{11}$$

Therefore, the plausible values of a and b are

$$(a, b) = (1, -2), \left(\frac{32}{5}, \frac{158}{11}\right).$$

(b) Let $p, q, r \in \mathbb{R}$ with $p, q, r > 0$. Consider the matrix B defined by:

$$B = \begin{pmatrix} 1 & 1 & 1 \\ p & q & r \\ p^3 & q^3 & r^3 \end{pmatrix}.$$

Show that $\det(B) \neq 0$ when p, q, r are pairwise distinct.
Hint: A set of values is pairwise distinct if no two values are equal to each other.

Solution: Direct computation of the determinant gives:

$$\begin{aligned} \det(B) &= 1 \begin{vmatrix} q & r \\ q^3 & r^3 \end{vmatrix} - 1 \begin{vmatrix} p & r \\ p^3 & r^3 \end{vmatrix} + 1 \begin{vmatrix} p & q \\ p^3 & q^3 \end{vmatrix} \\ &= (qr^3 - q^3r) - (pr^3 - p^3r) + (pq^3 - p^3q) \\ &= qr(r^2 - q^2) - pr(r^2 - p^2) + pq(q^2 - p^2) \\ &= (q-p)(r-p)(r-q)(p+q+r). \end{aligned}$$

Since $p, q, r \in \mathbb{R}^+$ are pairwise distinct, each of $(q-p), (r-p), (r-q)$ is non-zero. Also, we have $(p+q+r) > 0$. Therefore $\det(B)$ is non-zero.

10 marks 2. A vector $\vec{v} \in \mathbb{R}^n$ is said to be *balanced* if the entries of $\vec{v}$ sum to zero.

(a) Determine (with justifications) which of the following vector(s) are balanced vectors.

$$\vec{p} = \begin{pmatrix} 1 \\ 3 \\ -5 \\ 2 \end{pmatrix} \qquad \vec{q} = \begin{pmatrix} 2 \\ 5 \\ -1 \\ 0 \\ -6 \end{pmatrix} \qquad \vec{r} = \begin{pmatrix} \sqrt{8} \\ \sqrt{2} \\ -\sqrt{18} \end{pmatrix}$$

Solution: We can check the sum of entries for each vector:

- $\vec{p}$ is not balanced because $1 + 3 - 5 + 2 \neq 0$.
- $\vec{q}$ is balanced since $2 + 5 - 1 + 0 - 6 = 0$.
- $\vec{r}$ is balanced since $\sqrt{8} + \sqrt{2} - \sqrt{18} = 2\sqrt{2} + \sqrt{2} - 3\sqrt{2} = 0$.

(b) Let V_n be the set of all balanced vectors from $\mathbb{R}^n$ for $n \geq 2$. Let $\vec{u} = (u_1, u_2, \cdots, u_n)^\top$ and $\vec{w} = (w_1, w_2, \cdots, w_n)^\top$ be two distinct (fixed but unknown) vectors from V_n.

Determine (with justifications) whether the following vectors must be in the set V_n:

i. The vector $\vec{a} = \vec{u} + \vec{w}$

Solution: Since $\vec{u}, \vec{w} \in V_n$, we have that $u_1 + u_2 + \cdots + u_n = w_1 + w_2 + \cdots + w_n = 0$. The vector is $\vec{a} = \vec{u} + \vec{w} = (u_1 + w_1, u_2 + w_2, \ldots, u_n + w_n)$. Since

$$(u_1 + w_1) + (u_2 + w_2) + \cdots + (u_n + w_n) = (u_1 + u_2 + \cdots + u_n) + (w_1 + w_2 + \cdots + w_n) = 0$$

it follows that $\vec{a} = \vec{u} + \vec{w} \in V_n$.

ii. The vector $\vec{b} = (\lfloor u_1 \rfloor, \lfloor u_2 \rfloor, \cdots, \lfloor u_n \rfloor)^\top$, where $\lfloor x \rfloor$ is the floor function (the largest integer n such that $n \leq x$).

Solution: We will provide counterexamples to show that there is at least one vector $\vec{u} \in V_n$ where its transformed vector $\vec{b} \notin V_n$.

For this, we note the following:

$$\lfloor 0.5 \rfloor = 0 \qquad \lfloor -0.5 \rfloor = -1 \qquad \lfloor 0 \rfloor = 0$$

So if we choose $\vec{u} = (0.5, -0.5, 0, \ldots 0)^\top$, we have $\vec{u} \in V_n$. But $\vec{b} = (0, -1, 0, \ldots, 0)^\top \notin V_n$ since the entries do not sum to zero.

iii. The vector $\vec{c} = A\vec{u}$, where A is defined as follows:

$$A = \begin{pmatrix} a_1 & a_n & \cdots & a_3 & a_2 \\ a_2 & a_1 & a_n & & a_3 \\ \vdots & a_2 & a_1 & \ddots & \vdots \\ a_{n-1} & & \ddots & \ddots & a_n \\ a_n & a_{n-1} & \cdots & a_2 & a_1 \end{pmatrix}$$

Note: This matrix A is known as a circulant matrix

Solution: The transformed vector is

$$\vec{c} = A\vec{u} = \begin{pmatrix} a_1u_1 + a_nu_2 + \cdots + a_2u_n \\ a_2u_1 + a_1u_2 + \cdots + a_3u_n \\ \vdots \\ a_nu_1 + a_{n-1}u_2 + \cdots + a_1u_n \end{pmatrix}$$

The sum of elements of $\vec{c}$ is

$$\begin{aligned} &(a_1 + a_2 + \cdots + a_n)u_1 + (a_n + a_1 + \cdots + a_{n-1})u_2 + \cdots + (a_2 + a_3 + \cdots a_1)u_n \\ =&(a_1 + a_2 + \cdots + a_n)u_1 + (a_1 + a_2 + \cdots + a_n)u_2 + \cdots + (a_1 + a_2 + \cdots + a_n)u_n \\ =&(a_1 + a_2 + \cdots + a_n)(u_1 + u_2 + \cdots + u_n) \\ =&0 \end{aligned}$$

it follows that $\vec{c} = A\vec{u} \in V_n$.

(c) Given three vectors $\vec{v}_1, \vec{v}_2, \vec{v}_3 \in V_3$.

i. Show that $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ cannot be a basis of $\mathbb{R}^3$.

Solution: If we can show that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ can never be linearly independent, then it cannot be a basis (definition 2.22). Since we know the sum of entries must be zero, define the vectors as follows: $\vec{v}_1 = (a, b, -a-b)^\top, \vec{v}_2 = (c, d, -c-d)^\top, \vec{v}_3 = (e, f, -e-f)^\top$. Solving $\alpha\vec{v}_1 + \beta\vec{v}_2 + \gamma\vec{v}_3 = \vec{0}$ for $\alpha, \beta, \gamma \in \mathbb{R}$ is equivalent to solving the system

$$\left(\begin{array}{ccc|c} a & c & e & 0 \\ b & d & f & 0 \\ -a-b & -c-d & -e-f & 0 \end{array}\right)$$

It is clear that $R_3 = -(R_1 + R_2)$, rendering the last row all zeros. There is one free variable and hence the system has infinitely many solutions. Hence, $\vec{v}_1, \vec{v}_2, \vec{v}_3$ are not linearly independent. It follows that $\vec{v}_1, \vec{v}_2, \vec{v}_3$ cannot form a basis of $\mathbb{R}^3$.

Solution: (Alternate solution): We can also show that the set of vectors is not a spanning set of $\mathbb{R}^3$. Let $\vec{w} \in \text{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$, then $\vec{w} = \alpha\vec{v}_1 + \beta\vec{v}_2 + \gamma\vec{v}_3$ for some

$\alpha, \beta, \gamma \in \mathbb{R}$. In particular, the sum of the entries in $\vec{w}$ is zero (from previous part). Therefore, the vector $(1,0,0)^\top \notin \text{span}(\vec{v}_1, \vec{v}_2, \vec{v}_3)$. Hence it is not a spanning set (Definition 2.22)

ii. Describe all possible geometric objects (lines, planes, etc.) represented by $\text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$. Provide appropriate justification and/or numeric examples for your answers.

Solution: The $\text{span}\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ can be either the zero vector $\{\vec{0}\}$ (origin), a line through the origin, or a plane that contains the origin.

- **Case 1:** The zero vector $\{\vec{0}\}$ (origin): If $\vec{v}_1 = \vec{v}_2 = \vec{v}_3 = \vec{0}$, then the span of all three vectors is just $\{\vec{0}\}$.
- **Case 2:** A line through the origin: Let $\vec{v}_1 \neq \vec{0}$. We can choose $\vec{v}_2 = k\vec{v}_1$ and $\vec{v}_3 = \ell\vec{v}_1$ for some scalars k and ℓ, so vectors in the span of all three have the form:

  $$\lambda_1\vec{v}_1 + \lambda_2\vec{v}_2 + \lambda_3\vec{v}_3 = \lambda_1\vec{v}_1 + \lambda_2 k\vec{v}_1 + \lambda_3\ell\vec{v}_1 = (\lambda_1 + \lambda_2 k + \lambda_3\ell)\vec{v}_1$$

  implying that the span of the vectors is a straight line through the origin with direction vector $\vec{v}_1$.
- **Case 3:** A plane through the origin: Let $\vec{v}_1 \neq \vec{0}$, we choose $\vec{v}_2 \neq k\vec{v}_1$ for any real number k. From part (i), we know that $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ must be linearly dependent, i.e., $\vec{v}_3$ can always be written as some linear combination of $\vec{v}_1$ and $\vec{v}_2$. Hence, we can form a basis for V_3 spanned by $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ by removing $\vec{v}_3$ (e.g Tutorial 2 Q8). Hence $\text{span}(\vec{v}_1, \vec{v}_2) = \{t\vec{v}_1 + s\vec{v}_2 | t, s \in \mathbb{R}\}$ which is a plane that passes through the origin.

[*Other arguments should be accepted while they make sense. e.g. providing explicit examples and showing that have the desired properties*]

Note: The dot product $\vec{v} \cdot \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} = 0$ for all $\vec{v} \in V$. This implies that $(1,1,1)^\top$ is a normal vector to the plane spanned by $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$, and the equation of the plane is $x + y + z = 0$.

11 marks 3. An $N \times N$ matrix A is said to be a *generating matrix* if there exists a vector $\vec{v} \in \mathbb{R}^N$ such that:

$$\text{span}\left(\vec{v}, A\vec{v}, A^2\vec{v}, A^3\vec{v}, \ldots, A^k\vec{v}, \ldots\right) = \mathbb{R}^N$$

(a) Determine (with justification) if the matrix C is a generating matrix.

$$C = \begin{pmatrix} 1 & 1 & 1 \\ 1 & 1 & 0 \\ 1 & 0 & 2 \end{pmatrix}$$

Solution: For this one, we want to show that C is a generating matrix. To do that, we will need a vector $\vec{v}$ and check that it meets the required conditions. There are infinitely many choices for vector $\vec{v}$, but for this solution, we'll use $\vec{v} = (1, 0, 0)^\top$ as the example. We first compute the first few terms of the series.

$$\vec{v} = \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} \qquad C\vec{v} = \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} \qquad C^2\vec{v} = \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix}$$

As sets, we have the following inequalities:

$$\text{span}\left(\vec{v}, C\vec{v}, C^2\vec{v}\right) \subseteq \text{span}\left(\vec{v}, C\vec{v}, C^2\vec{v}, C^3\vec{v}, \ldots, C^k\vec{v}, \ldots\right) \subseteq \mathbb{R}^N$$

The first inequality is because increasing the number of vectors in a set cannot decrease the span. The second inequality follows from the fact that every possible span is a subset of $\mathbb{R}^3$. If we can show that $\text{span}\left(\vec{v}, C\vec{v}, C^2\vec{v}\right) = \mathbb{R}^3$, then the inequality gives us the required condition for C to be a generating matrix.
We will show that $\vec{v}, C\vec{v}, C^2\vec{v}$ are linearly independent. Using the linearly independent condition, we want to find solutions to:

$$\alpha \begin{pmatrix} 1 \\ 0 \\ 0 \end{pmatrix} + \beta \begin{pmatrix} 1 \\ 1 \\ 1 \end{pmatrix} + \gamma \begin{pmatrix} 3 \\ 2 \\ 3 \end{pmatrix} = \vec{0}$$

This gives the augmented matrix:

$$\left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 1 & 3 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 1 & 3 & 0 \\ 0 & 1 & 2 & 0 \\ 0 & 0 & 1 & 0 \end{array}\right) \qquad R_3 \to R_3 - R_2$$

With a full set of pivots, we can back substitute to get the unique solution $\alpha = \beta = \gamma = 0$. Therefore, we conclude that $\{\vec{v}, C\vec{v}, C^2\vec{v}\}$ forms a set of three linearly independent vectors. By Theorem 2.24, we have this set forms a basis and hence

$$\text{span}\left(\vec{v}, C\vec{v}, C^2\vec{v}\right) = \mathbb{R}^3$$

Therefore C is a generating matrix.

(b) Determine (with justification) if the matrix B is a generating matrix.

$$B = \begin{pmatrix} 3 & -4 & 2 \\ 3 & -5 & 3 \\ 3 & -6 & 4 \end{pmatrix}$$

Solution: We will show that this matrix is NOT a generating matrix. To show this, we need to show that there exists no vector $\vec{v}$ such that

$$\text{span}\left(\vec{v}, B\vec{v}, B^2\vec{v}, B^3\vec{v}, \ldots, B^k\vec{v}, \ldots\right) = \mathbb{R}^N$$

For that, we check the powers of B

$$B^2 = B \cdot B = \begin{pmatrix} 3 & -4 & 2 \\ 3 & -5 & 3 \\ 3 & -6 & 4 \end{pmatrix} \cdot \begin{pmatrix} 3 & -4 & 2 \\ 3 & -5 & 3 \\ 3 & -6 & 4 \end{pmatrix} = \begin{pmatrix} 3 & -4 & 2 \\ 3 & -5 & 3 \\ 3 & -6 & 4 \end{pmatrix} = B$$

Since we have $B^2 = B$, we also have for each integer power of B:

$$\begin{aligned} B^3 &= (B^2)B = (B)B = B^2 = B \\ B^4 &= (B^3)B = (B)B = B^2 = B \\ &\vdots \\ B^k &= (B^{k-1})B = (B)B = B^2 = B \end{aligned}$$

Hence, for every vector $\vec{v}$, we have:

$$B\vec{v} = B^2\vec{v} = B^3\vec{v} = B^4\vec{v} = \cdots = B^k\vec{v}$$

Which means:

$$\left\{\vec{v}, B\vec{v}, B^2\vec{v}, B^3\vec{v}, \ldots, B^k\vec{v}, \ldots\right\} = \{\vec{v}, B\vec{v}\}$$

Therefore:

$$\text{span}\left(\vec{v}, B\vec{v}, B^2\vec{v}, B^3\vec{v}, \ldots, B^k\vec{v}, \ldots\right) = \text{span}\left(\vec{v}, B\vec{v}\right)$$

Since there are at most two vectors in the set. Proposition 2.16 tells us that this cannot be a spanning set of $\mathbb{R}^N$. Hence

$$\text{span}\left(\vec{v}, B\vec{v}\right) \neq \mathbb{R}^N$$

Note: We can show that this is a subset of $\mathbb{R}^N$ as opposed to just not equal. But this is sufficient for the purpose of this question.
Since there exists no vectors for which the span gives us the full $\mathbb{R}^3$, we conclude the B cannot be a generating matrix.

(c) For this part, we only consider 2×2 matrices. Let I be the identity matrix and define the set $D = \{kI \mid k \in \mathbb{R}\}$ to be all matrices that are a scalar multiple of the identity matrix. Prove the following statement:

The 2×2 matrix M is a generating matrix if and only if $M \notin D$.

Solution: For this question, we have to show both implications. One way to do this is:

1. Show if M is a generating matrix then $M \notin D$
2. Show if M is NOT a generating matrix then $M \in D$

For the first statement: Suppose M is a generating matrix. Then there exists $\vec{v}$ such that $\text{span}(\vec{v}, M\vec{v}) = \mathbb{R}^2$. Since we have two vectors spanning $\mathbb{R}^2$, Theorem 2.24 says that $\vec{v}, M\vec{v}$ are linearly independent. That is: $M\vec{v} \notin \text{span}(\vec{v})$ and $M\vec{v} \neq \lambda\vec{v}$ for any value of λ (e.g. $M\vec{v}$ is not a scalar multiple of $\vec{v}$). If $M \in D$, then $M\vec{v} = kI\vec{v} = k\vec{v}$, which is a scalar multiple of $\vec{v}$ and hence gives a contradiction. Therefore $M \notin D$.
Note: The choice $\vec{v}, M\vec{v}$ is sufficient to be a spanning set. Otherwise, $M\vec{v} \in \text{span}(\vec{v})$, which means $M\vec{v} = \alpha\vec{v}$ for some constant α. Then all higher powers $M^k\vec{v} \in \text{span}(\vec{v})$, which contradicts M being a generating matrix.

For the second statement: Suppose M is not a generating matrix. Then there cannot exists a vector $\vec{v}$ such that $\text{span}(\vec{v}, M\vec{v}) = \mathbb{R}^2$. That is, for any vector $\vec{v}$, we have $\text{span}(\vec{v}, M\vec{v}) \subset \mathbb{R}^2$ (strict subset). This case only happens when $M\vec{v} \in \text{span}(\vec{v})$. Otherwise, they are linearly independent and forms a basis. So for any vector $\vec{v}$ we must have $M\vec{v} = k\vec{v}$ for some k. We can write out the matrix equation as follows:

$$\begin{pmatrix} a & b \\ c & d \end{pmatrix} \begin{pmatrix} v_1 \\ v_2 \end{pmatrix} = k \begin{pmatrix} v_1 \\ v_2 \end{pmatrix}$$

Equating the two coordinates gives:

$$a \cdot v_1 + b \cdot v_2 = k \cdot v_1$$
$$c \cdot v_1 + d \cdot v_2 = k \cdot v_2$$

Since this is true for all v_1, v_2, the first equation gives $a = k, b = 0$ and the second equation gives $c = 0, d = k$. Combining, this gives

$$M = \begin{pmatrix} k & 0 \\ 0 & k \end{pmatrix} = kI \in D$$

which is the required conclusion.

11 marks

4. In machine learning, a *classifier* is a model that is used to perform classification tasks. Given some input $\vec{x}$, the model determines which category y it belongs to. For simplicity, we will initially consider only two outcomes. That is, $y \in \{-1, 1\}$. In the context of telecommunication contracts, we can think of:

   - The vector $\vec{x}$ as a list of personal information for a customer, which may include contract length, monthly charges, account age, etc.
   - The label y for whether the customer is likely to discontinue their subscription (called a churn). In this case, we denote $y = 1$ that the customer churns and $y = -1$ that the customer does not churn.

   The decision rule of a classifier is a function $D(\vec{x}) = h(\vec{w}^\top \vec{x})$, where $\vec{w}^\top = (w_1, w_2, \ldots, w_n)$ is a weight vector, and $h(\cdot)$ is a user-specified activation function. The classifier *learns* through adjusting the weight vector $\vec{w}$ to minimise its classification error. This can be accomplished via feeding the model a set of training data $\{\vec{x}_1, \vec{x}_2, \ldots, \vec{x}_m\} \in \mathbb{R}^n$ for which the labels $\{y_1, y_2, \ldots, y_m\}$ are known. The function h is typically chosen as a function that returns only two possible values, e.g., the heaviside function. Figure 1 depicts how a classifier works:

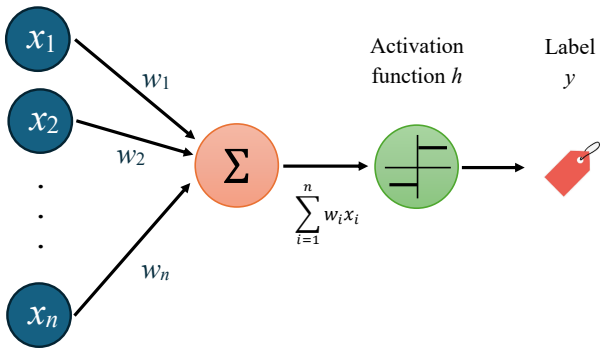


Figure 1: An illustration of a classifier taking a vector $\vec{x} = (x_1, \ldots, x_n)^\top$ as input and returning a real number y as an output.

For this question, we will use the following training data points and their corresponding labels:

i	$\vec{x}_i$	y_i
1	$(1, 2, 1, 0)^\top$	1
2	$(-1, 2, 3, 1)^\top$	-1
3	$(1, 0, 5, 2)^\top$	1
4	$(1, 1, 3, 1)^\top$	1

(a) Consider the following function h used in the decision rule:

$$h(x) = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{otherwise} \end{cases}$$

Three machine learning researchers Jake, Maryam, and Qing have proposed using three different sets of weights for classifying the training data. Jake proposed setting $\vec{w} =$

$(2,-1,1,-2)^\top$, Maryam proposed setting $\vec{w} = (4,-2,1,-3)^\top$, and Qing proposed setting $\vec{w} = (1,-2,2,-3)^\top$.

i. Determine the accuracy of the predictions by using the weight vectors proposed by Jake, Maryam, and Qing to compute the labels of the training data points. That is, calculate $h(\vec{w}^\top \vec{x_1}), h(\vec{w}^\top \vec{x_2}), h(\vec{w}^\top \vec{x_3})$, and $h(\vec{w}^\top \vec{x_4})$ for each of the weight vectors and explain how the result matches with the known outputs of $(y_1, y_2, y_3, y_4) = (1,-1,1,1)$.

Solution: For the sake of compactness, we evaluate the predicted labels using matrix products. Let $X = (\vec{x}_1|\vec{x}_2|\vec{x}_3|\vec{x}_4)$, and define the following map $H(z_1,z_2,z_3,z_4) = (h(z_1), h(z_2), h(z_3), h(z_4))$.

- For Jake,

$$\vec{w}^\top X = (2,-1,1,-2)\begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 3 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix} = (1,-3,3,2)$$

  The predicted labels using Jake's model are $H(1,-3,3,2) = (1,-1,1,1)$.

- For Maryam,

$$\vec{w}^\top X = (4,-2,1,-3)\begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 3 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix} = (1,-8,3,2)$$

  The predicted labels using Maryam's model are $H(1,-8,3,2) = (1,-1,1,1)$.

- For Qing,

$$\vec{w}^\top X = (1,-2,2,-3)\begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 3 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix} = (-1,-2,5,2)$$

  The predicted labels using Jake's model are $H(-1,-2,5,2) = (-1,-1,1,1)$.

Based on these results, Jake's and Maryam's models accurately predict the labels for all four training data points, whilst Qing's does not.

ii. Using your results from the previous part, show that there is an infinite number of possible weight vectors $\vec{w}$ that can correctly classify all four training data points.

Solution: From part (i), we know that Jake's weight vector will correctly classify all four training data points. We can take a scalar multiple of the weight vector, i.e., $\vec{\omega} = \lambda(2, -1, 1, -2)^\top$. By multiplying with the matrix of training data, we have:

$$\vec{\omega}^\top X = \lambda(2, -1, 1, -2) \begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 3 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix} = (\lambda, -3\lambda, 3\lambda, 2\lambda)$$

For this to match the actual labels we need $H(\lambda, -3\lambda, 3\lambda, 2\lambda) = (1, -1, 1, 1)$. This is satisfied when $\lambda > 0$, hence giving *infinitely many* weight vectors that will correctly classify all four training data points.

Solution: (Alternate solution): From part (i), there are two weight vectors that will correctly classify all four training data points. We want to create an infinite number of weights based on these two points. We will now mimic the proof in Tutorial 1 Q5(a). Let the weight vectors guessed by Jake and Maryam be $\vec{w}_1$ and $\vec{w}_2$ respectively. Define $\vec{\omega} = \lambda\vec{w}_1 + (1 - \lambda)\vec{w}_2$ for some real paramter λ. Then

$$\vec{\omega} = \begin{pmatrix} 2\lambda + 4(1 - \lambda) \\ -\lambda - 2(1 - \lambda) \\ \lambda + (1 - \lambda) \\ -2\lambda - 3(1 - \lambda) \end{pmatrix} = \begin{pmatrix} -2\lambda + 4 \\ \lambda - 2 \\ 1 \\ \lambda - 3 \end{pmatrix}$$

By multiplying with the matrix of training data, we have:

$$\vec{\omega}^\top X = (-2\lambda + 4, \lambda - 2, 1, \lambda - 3) \begin{pmatrix} 1 & -1 & 1 & 1 \\ 2 & 2 & 0 & 1 \\ 1 & 3 & 5 & 3 \\ 0 & 1 & 2 & 1 \end{pmatrix}$$

$$= (1, 5\lambda - 8, 3, 2)$$

For this to match the actual labels, we need $H(1, 5\lambda - 8, 3, 2) = (1, -1, 1, 1)$. This is satisfied when $\lambda \leq 8/5$, hence giving *infinitely many* weight vectors that will correctly classify all four training data points.

(b) For this part, consider the new function $h(x) = x$, i.e. the identity function. To allow for better flexibility of the classifier, the decision rule is now modified to be

$$D(\vec{x}) = h(\vec{w}^\top \vec{x} + b)$$

where $b \in \mathbb{R}$ is a bias term.

i. By taking $b = -1$, determine the set of weight vector(s) that will fit all the four training data points.

Solution: To determine the set of feasible weights, we will set up the equations satisfied by the weight vector $\vec{w}^\top = (w_1, w_2, w_3, w_4)$:

$$\begin{aligned} w_1 + 2w_2 + w_3 + 0w_4 - 1 &= 1 \\ -w_1 + 2w_2 + 3w_3 + w_4 - 1 &= -1 \\ w_1 + 0w_2 + 5w_3 + 2w_4 - 1 &= 1 \\ w_1 + w_2 + 3w_3 + w_4 - 1 &= 1 \end{aligned}$$

By simplifying the linear equations and putting them in an augmented matrix, we have

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 2 \\ -1 & 2 & 3 & 1 & 0 \\ 1 & 0 & 5 & 2 & 2 \\ 1 & 1 & 3 & 1 & 2 \end{array}\right)$$

By performing Gaussian elimination, we have

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 2 \\ -1 & 2 & 3 & 1 & 0 \\ 1 & 0 & 5 & 2 & 2 \\ 1 & 1 & 3 & 1 & 2 \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 2 \\ 0 & 4 & 4 & 1 & 2 \\ 0 & -2 & 4 & 2 & 0 \\ 0 & -1 & 2 & 1 & 0 \end{array}\right) \qquad \begin{aligned} R_2 &\to R_2 + R_1 \\ R_3 &\to R_3 - R_1 \\ R_4 &\to R_4 - R_1 \end{aligned}$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 2 \\ 0 & 4 & 4 & 1 & 2 \\ 0 & 0 & 12 & 5 & 2 \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \qquad \begin{aligned} R_3 &\to 2R_3 + R_2 \\ R_4 &\to 2R_4 - R_3 \end{aligned}$$

There are infinitely many weight vectors possible for the system, since there is a free variable w_4. By taking $w_4 = \alpha \in \mathbb{R}$, it is possible to write the solution of the system as

$$\begin{aligned} 12w_3 + 5w_4 = 2 &\implies w_3 = \frac{2 - 5\alpha}{12} \\ 4w_2 + 4w_3 + w_4 = 2 &\implies w_2 = \frac{1}{4}\left(2 - \alpha - \frac{2 - 5\alpha}{3}\right) = \frac{2 + \alpha}{6} \\ w_1 + 2w_2 + w_3 = 2 &\implies w_1 = 2 - \frac{2 - 5\alpha}{12} - \frac{2 + \alpha}{3} = \frac{\alpha + 14}{12}. \end{aligned}$$

The general form of the weight vector can be written as

$$\vec{w} = \frac{1}{6}\begin{pmatrix} 7 \\ 2 \\ 1 \\ 0 \end{pmatrix} + \frac{\alpha}{12}\begin{pmatrix} 1 \\ 2 \\ -5 \\ 12 \end{pmatrix}, \qquad \alpha \in \mathbb{R}$$

ii. Using the previous part or otherwise, determine the set of values $b \in \mathbb{R}$ such that there exists a weight vector to correctly classify all the training data points.

Solution: It is easy to see that the augmented matrix will now be of the form

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1-b \\ -1 & 2 & 3 & 1 & -1-b \\ 1 & 0 & 5 & 2 & 1-b \\ 1 & 1 & 3 & 1 & 1-b \end{array}\right)$$

Using the same row operations as before, we get:

$$\left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1-b \\ -1 & 2 & 3 & 1 & -1-b \\ 1 & 0 & 5 & 2 & 1-b \\ 1 & 1 & 3 & 1 & 1-b \end{array}\right) \sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1-b \\ 0 & 4 & 4 & 1 & -2b \\ 0 & -2 & 4 & 2 & 0 \\ 0 & -1 & 2 & 1 & 0 \end{array}\right) \begin{array}{l} \\ R_2 \to R_2 + R_1 \\ R_3 \to R_3 - R_1 \\ R_4 \to R_4 - R_1 \end{array}$$

$$\sim \left(\begin{array}{cccc|c} 1 & 2 & 1 & 0 & 1-b \\ 0 & 4 & 4 & 1 & -2b \\ 0 & 0 & 12 & 5 & -2b \\ 0 & 0 & 0 & 0 & 0 \end{array}\right) \begin{array}{l} \\ \\ R_3 \to 2R_3 + R_2 \\ R_4 \to 2R_4 - R_3 \end{array}$$

By performing row operations we will certainly have a final row with all zeros, and hence the system will always be consistent for all values of b. Hence, a set of weight vectors that will fit the training data can be found for all $b \in \mathbb{R}$.